Andrew W. Shenstone

Shenstone, Andrew - Application for Senior Front-end Engineer @ Tenna

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SUMMARY

Senior software engineer with 12+ years building customer-facing and internal products where the UI has to stay legible under real-world complexity—multi-step workflows, dense data, and performance that holds up for daily operators. I ship primarily in React and TypeScript with Node.js in the loop, care about reusable components and design-system-style consistency, and treat testing (Vitest, Cypress) and code review as part of the product bar. I've partnered closely with design and product at Bloom and Tilia, owned features from prototype thinking through production, and optimized frontends for maintainability and speed. Tenna's mission—connecting construction assets, people, and data—fits that pattern: high-trust B2B software, field-adjacent realities, and interfaces that have to earn trust on every job. My AWS footprint today centers on S3 and EC2-class usage; I'm comfortable deepening exposure to the broader AWS services your stack relies on.

SKILLS

Frontend: TypeScript, JavaScript, React, Next.js, Node.js, Component architecture & reusable UI, Responsive layout & UX polish, Performance optimization, Accessibility-minded delivery

Testing & quality: Vitest, Cypress, Unit, integration, and E2E testing, Code review & maintainability

Collaboration & delivery: Cross-functional work with design, product, and backend, Agile / iterative delivery, Documentation for onboarding and shared context, Pairing and informal technical lead (no direct reports)

Architecture & async: Monorepos, Async workflows (Inngest, webhooks, background jobs), ETL pipelines, Performance optimization

Cloud & Tools: AWS (S3, EC2), Vercel, Algolia, Inngest, Clerk Auth, Stripe, Git, CI/CD, pnpm, Turbo, Cursor IDE (Top 1% user)

EDUCATION

B.S. Finance, Santa Clara University — 2010

EXPERIENCE

Founder & Lead Engineer

Plantfolio | November 2023 - Present

Product Leadership & Architecture

- Architected and built a comprehensive plant management platform from the ground up, serving as both technical lead and product owner
- Owned the full feature cycle: joined customer calls to identify friction points, architected solutions, implemented, tested, and verified success in production
- Designed and implemented a sophisticated monorepo architecture with 12+ packages, enabling rapid development and maintainable codebases
- Led product strategy and user experience decisions, making product micro-decisions and navigating ambiguity to deliver value iteratively

Technical Achievements

- AI Integration: Built advanced AI-powered features including plant identification from photos, contextual chatbot (Sage), and smart text extraction from plant tags using OpenAI's vision and language models
- Modern Frontend Architecture: Implemented Next.js with React, React Server Components, and TypeScript, achieving exceptional performance and developer experience

- Complex Data Modeling: Designed and implemented sophisticated PostgreSQL schemas with Prisma, handling complex relationships between users, plants, folios, and media with 15+ interconnected models
- Supabase to Neon: Shipped on Supabase (PostgreSQL with Row-Level Security policies for multi-tenant isolation) before migrating the data layer to Neon; strong hands-on experience with RLS and Postgres-centric access control
- Content Management: Created sophisticated content management system with rich text editing, plant tagging, and flexible data structures

Scale & Performance

- Managed a database of 40,000+ plant species with advanced search capabilities using Algolia and vector embeddings
- Implemented comprehensive media management system with AWS S3, supporting image optimization, galleries, and metadata extraction
- Built robust authentication and authorization systems with Clerk, supporting multiple user types and subscription tiers
- Optimized application performance through React Server Components, efficient data fetching, and strategic caching

Client Case Study - AI Signage

- Delivered a production rollout for an independent garden center to automate plant signage generation; one independent retail customer in ongoing production who has been an enthusiastic reference
- Enabled bulk plant import (plain text) to printable signs in seconds; reduced a weekend-long manual process to minutes after initial setup
- Demonstrated ability to translate customer requirements into scalable technical solutions and shorten feedback loops

Full-Stack Ownership

- Developed complete payment integration with Stripe, subscription management, and user onboarding flows
- Built comprehensive email system with React Email, supporting transactional emails, notifications, and marketing campaigns

- Implemented advanced search and filtering capabilities with Algolia, supporting complex plant queries and user preferences

Code Quality & Leadership

- Established coding standards and architectural patterns that enabled rapid, consistent development
- Implemented comprehensive testing strategy with Vitest and Cypress, maintaining high code quality
- Built reusable component library and design system, ensuring consistent UI/UX across the platform

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Senior Software Engineer

Tilia | January 2023 - August 2023

- Partnered with cross-functional teams in product and design to revamp critical user-facing features
- Built and shipped work in support of KYC, identity verification, and risk-adjacent tooling as part of Tilia's payments platform (tools team context)
- Spearheaded and guided a team through the successful adoption of Chakra UI, reducing bespoke components and establishing cohesive design patterns
- Mentored engineers through pairing and code review in an informal technical lead capacity (no direct reports)
- Maintained exceptional unit test coverage of over 90% for the Tools team application, alongside comprehensive end-to-end testing
- Led the swift creation and deployment of a homepage reporting dashboard within the initial two months of joining
- Role ended in a company-wide layoff affecting approximately one-third of employees

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Senior Software Engineer

Bloom.wine | September 2020 - October 2022

- Architected a greenfield Next.js application for the primary admin interface of the Bloom product; much of that stack remains in production across many winery clients today
- Engineered Ruby on Rails GraphQL API endpoints shared by multiple applications across mobile and web devices
- Mentored engineers through pairing and code review in an informal technical lead capacity (no direct reports)
- Collaborated with designers using Figma, establishing component library standards usable by both engineering and design
- Integrated DatoCMS and extended application functionality through third-party API integrations (Shopify, Mailchimp, Stripe, Shipstation, ActiveCampaign); all Bloom winery clients were on Shopify Plus

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Software Engineer II

Tune Inc - Marketplace Automation | August 2019 - May 2020

- Built and operated ETL-style data pipelines and integrations supporting marketplace automation workflows
- Automated a client workflow using a Node.js/Express API and React.js, reducing the time to complete the task from 10-15 minutes to 10-30 seconds
- Modernized the primary legacy React.js application by moving from a custom Webpack process to react-scripts, resulting in a 50% reduction in production build size and a 60% reduction in NPM dependencies
- Reduced development machine CPU usage from 75% to <5% by simplifying the build pipeline and dependency graph
- Role ended in a company-wide layoff

Software Developer

Ply Interactive Inc | June 2014 - July 2019

React Native

- Shipped multiple production React Native applications for agency clients—cross-platform mobile apps delivered end to end
- Washington State Fair: React Native app in sustained multi-year production; very high traffic during fair season, from tens of thousands of users into the hundreds of thousands cumulatively over seasons

Web & platforms

- Developed numerous web applications including Georgetownbeer.com, WashingtonWine.org, TwoWolvesWine.com, Novaerus.com, WellAir, and Plasma-air.com
- Built a standalone Ruby gem Content Management System, enabling simplified content management within Ruby on Rails projects
- Transitioned the agency to modern technologies like GraphQL, Gatsby.js, and React.js, improving delivery times for new projects
- Bloom.wine Product Development: Built extensive reporting features using complex PostgreSQL queries and interactive filtering, saving clients thousands of hours on previously manual tasks
- Engineered a flexible Wine Club membership structure for wineries, allowing for diverse client configuration and broadening the range of potential sales targets
- Communicated and enforced the standards of Shopify's Polaris React.js component library with designers, resulting in efficiencies gained from leveraging pre-built components

COMPANY REFERENCE

Plantfolio (2023-Present) - AI-powered plant management platform with 40,000+ plant database serving both consumer garden management and independent garden center services - plantfolio.garden/

Tilia (2023) - All-in-one payment platform for digital economies - tilia.io/

Bloom.wine (2020-2022) - Commerce services for the alcohol industry; all clients on Shopify Plus - bloom.wine/

Tune Inc - Marketplace Automation (2019-2020) - SaaS platform for managing marketing partnerships across mobile and web - tune.com/

Ply Interactive Inc (2014-2019) - Boutique agency: web and production React Native apps (e.g. Washington State Fair at major scale); developed Bloom.wine product internally (2017-2019)